Examples (WAVES AND MOTION)		
1.	Find the acceleration of particle performing simple harmonic motion (SHM) when it is at 0.6 m from its mean position. The time period of SHM is 0.05 sec. Also calculate maximum velocity if the amplitude of SHM is 2 m. **(Jun-2019)(Aug-2023)[NLJIET]**	**3**
	$x = 0.6\ m$ $A = 2\ m$ $T = 0.05\ s$ Acceleration $a = -\omega^2 x$ $= -\frac{4\pi^2}{T^2} x \qquad \omega = \frac{2\pi}{T}$ $= -\frac{4 \times (3.14)^2 \times 0.6}{(0.05)^2}$ $a = \cancel{x} - 9465.216\ m/s^2$ Maximum velocity $V_{max} = \omega A$ $= \frac{2\pi}{T} \cdot A$ $= \frac{2 \times 3.14 \times 2}{0.05}$ $V_{max} = 251.2\ m/s$ → Amplitude using maximum velocity $V_{max} = \omega A$ $A = \frac{V_{max}}{\omega}$ $= \frac{0.15}{2}$ $A = 7.5 \times 10^{-2}\ m$	

2.	A particle execute simple harmonic motion with amplitude of 0.1 m and a period of 0.5 s. Calculate its displacement, velocity and acceleration (1/12) s after it crosses the mean position. **(Jan-2024) [NLJIET]**	**3**

4. $A = 0.1\ m$

$T = 0.5\ s$

$t = 1/12\ s$

→ Angular frequency $(\omega) = \frac{2\pi}{T}$

$= \frac{2 \times 3.14}{0.5}$

$= 12.56$

→ Displacement $x = A \sin \omega t$

$= 0.1 \sin(12.56 \times 1/12)$

$= 0.1 \sin(1.046)$

$= 0.1\ (0.8654)$

$= 0.08654\ m$

→ Velocity $= \omega A \cos(\omega t)$

$= (12.56)(0.1) \cos((12.56) \times 1/12)$

$= 12.56 \times 0.1 \times \cos(1.046)$

$= 12.56 \times 0.1 \times 0.5016$

$= 0.6300\ m/s$

→ Acceration $a = -\omega^2 A \sin \omega t$

$= -(12.56)^2 (0.1)(0.8654)$

$= -13.6519\ m/s^2$

3.	An oscillator is performing SHM with an acceleration of 0.2 m/s^2 when displacement is 5 cm. Its maximum speed is 0.15 m/s. Find time period and amplitude of oscillator. **(Aug-2022) [NLJIET]**	**4**
	2. $a = 0.2\ m/s^2$ $x = 5cm = 0.05m$ $V_{max} = 0.15\ m/s$ $T = ?$ $A = ?$ → Find ω using accelaration equation $a = -\omega^2 x$ $\omega^2 = \frac{a}{x}$ $= \frac{0.2}{0.05}$ $\omega^2 = 4$ $\omega = 2\ rad/s$ $\omega = \frac{2\pi}{T}$ $T = \frac{2\pi}{\omega}$ $= \frac{2\pi}{2}$ $= \pi$ $T = 3.14\ Sec$	
4.	A particle of mass 0.50 kg executes a simple harmonic motion. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of the motion. (force constant k = 50 N/m). **(Jun-2025-New) [NLJIET]**	**4**

(force constant k = 50 N/m) ... amplitude of

Here, $m = 0.5$ kg

$V_{max} = 10$ m/s

$k = 50$ N/m

$A = ?$

$$v = \omega_0 \sqrt{A^2 - y^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}} = 10 \text{ rad/s}$$ (2 marks)

To obtain the amplitude

$$v_{max} = \omega_0 A$$

$$A = \frac{V_{max}}{\omega_0}$$

$$A = 1 \text{ m}$$ (2 marks)